

Function Point Analysis

1. Internal Logical Files (ILF)

Entity	RET	DET	FP
Users	1	11	10
Doctors	1	13	10
DoctorProfiles	1	4	7
PatientProfile	1	8	7
Appointments	1	8	7
AppointmentSlot	1	7	7
Availabilities	1	6	7

Chat	1	6	7
Messages	1	8	7
DoctorDocuments	1	7	7
Doctor_Patients_Relationship	1	7	7
Reviews	1	7	7
Prescriptions	1	10	10
Total FP (ILF)	-	-	110

2. External Interface Files (EIF)

No EIFs are present since all external entities are managed within the system.

Component	Function Points

Total FP (EIF)	0
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3. External Inputs (EI)

User Management

Entity	Operation	FTR	DET	FP
User	Register	2	8	4
User	Login	1	3	3
User	Update Profile	2	8	4
User	Delete Account	1	1	3

Doctor Management

Entity	Operation	FTR	DET	FP
Doctor	Create Profile	3	10	6
Doctor	Update Profile	3	10	6
Doctor	Upload Documents	2	5	4
Doctor	Set Availability	2	4	4
Doctor	Delete Profile	1	1	3

Appointment Management

Entity	Operation	FTR	DET	FP
Appointment	Book	3	6	6
Appointment	Reschedule	3	6	6

Appointment	Cancel	1	2	3
AppointmentSlot	Create	2	5	4
AppointmentSlot	Update	2	5	4

Communication

Entity	Operation	FTR	DET	FP
Chat	Initiate	2	4	4
Message	Send	2	4	4
Message	Mark as Read	1	2	3
Document	Upload	2	5	4
Prescription	Create	3	8	6

Review System

Entity	Operation	FTR	DET	FP
Review	Submit	2	5	4
Review	Update	2	5	4
Review	Delete	1	1	3

Admin Functions

Entity	Operation	FTR	DET	FP
User	Admin Modify	2	5	4
User	Admin Deactivate	1	2	3
Doctor	Verify Credentials	2	4	4
System	Monitor Performance	1	3	3

Total FP (EI)	115
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4. External Queries (EQ)

Category	FTR	DET	FP
Doctor Search	3	8	5
User Profile View	2	8	4
Appointment List (Patient)	2	6	4
Appointment List (Doctor)	2	6	4
Chat History	2	5	4
Document List	2	4	4
Prescription List	2	5	4

Review List	2	5	4
Doctor Reviews	1	4	3
Available Time Slots	2	4	4
System Health Check	1	2	3
User Management (Admin)	2	6	4
Total FP (EQ)			47

5. External Outputs (EO)

Category	FTR	DET	FP
Doctor Search Results	2	8	5
Appointment Confirmation	1	5	4

Prescription PDF	1	6	4
Document Download	1	3	4
Review Summary Report	2	5	5
System Metrics Dashboard	3	10	6
Total FP (EO)			28

Total Function Points Summary

Component	Function Points
ILF	110
EIF	0
EI	115

EQ	47
EO	28
TOTAL	300

Complexity Adjustment Factor (CAF)

Based on the system characteristics:

- Distributed architecture with microservices
- Real-time communication features
- File upload and document management
- Multiple user types and roles
- API Gateway integration
- Database complexity

Estimated CAF: 1.10

Adjusted Function Points

$$\text{AFP} = \text{FP} \times \text{CAF} = 300 \times 1.10 = \mathbf{330 \text{ Function Points}}$$